

Hands-On-Practice: Recommend Production Embedding Strategy for LegalTech

Description

This activity challenges you to make a high-stakes production decision by benchmarking multiple embedding models across competing dimensions. You'll evaluate accuracy, latency, and cost tradeoffs for a legal search system serving 500 law firms, then build a weighted decision framework that translates technical metrics into business impact. Through this hands-on practice, you'll develop essential skills in multi-dimensional model evaluation, total cost of ownership analysis, and executive-level technical communication that are critical for ML architects making production infrastructure decisions.

Learning Objectives

By the end of this activity, learners will be able to:

- Design and execute multi-dimensional benchmarking experiments measuring accuracy (recall@k), latency (p50/p95/p99), and throughput.
- Calculate total cost of ownership for embedding models at scale including compute, storage, and query serving costs.



- Build weighted decision matrices that explicitly trade off competing objectives based on business priorities.
- Translate technical benchmarks into business impact by connecting performance improvements to revenue protection and user satisfaction.



Assignment Components

Introductory Slide

Title: ML Architect Mission: Save \$1.2M ARR with the Right Model Choice

Subtitle: Balance 250ms latency crisis against accuracy requirements for 500 law firms

Tools to be used

Python 3.9+ with pandas and NumPy

Sentence-transformers (4 models: all-mpnet-base-v2, all-MiniLM-L6-v2, gte-small, e5-small-v2)

FAISS for vector search

time/timeit libraries for latency measurement

Excel or Google Sheets for cost modeling

Scenario

You are a machine learning architect at LegalTech Solutions, a SaaS platform that provides semantic search for legal documents serving 500 law firms with 50 million queries per month. Your current embedding system uses all-mpnet-base-v2 (768 dimensions), achieving 89% recall@10 but with 250ms average latency—painfully slow for lawyers expecting Google-speed results. Three clients representing \$1.2M in annual recurring revenue have paused contract renewals citing slow search performance.

Your CTO has allocated a \$5,000 monthly budget increase and asks you to evaluate alternatives that improve latency while maintaining acceptable accuracy. The business goal is to reduce average latency to under 100ms while keeping recall@10 above 85%. You must benchmark 4 models (current all-mpnet-base-v2 plus 3 alternatives: all-MiniLM-L6-v2, gte-small, e5-small-v2), project infrastructure costs at 50M+ monthly queries, and present an executive recommendation within one week.

Success means delivering a data-driven recommendation that balances latency, accuracy, and cost while quantifying business impact (churn reduction, revenue protection). Your recommendation will determine whether the company invests in infrastructure upgrades or pursues alternative solutions, affecting 500 law firms and \$12M in annual recurring revenue.

Instructions for Learners

Step 1: Multi-Model Accuracy Benchmarking

Download sample legal document corpus (10,000 documents) and 100 test queries with ground truth

For each of 4 models, generate embeddings for the corpus and test queries

Build FAISS indices for each model's embeddings

For each test query: search the index, retrieve top-10 results, calculate recall@5 and recall@10

Average recall@5 and recall@10 across all 100 queries for each model

Also calculate MRR (Mean Reciprocal Rank) for ranking quality assessment

Step 2: Latency and Throughput Benchmarking

For each model, measure query encoding latency on 100 test queries

Calculate p50 (median), p95, and p99 latency percentiles—not just averages

Measure FAISS search time separately (should be similar across models since index structure is same)

Calculate total latency = encoding time + search time

Measure throughput: queries per second each model can handle

Document embedding generation time for 2M documents (for one-time indexing cost)

Step 3: Cost Modeling and ROI Calculation

Calculate one-time costs: embedding generation for 2M documents (hours $\times$ EC2 c5.4xlarge pricing)

Calculate monthly recurring costs:

- Query serving compute (EC2 instances needed for 50M queries/month)
- Storage for vector index (S3 or EBS costs for storing $2M \times \text{dimension_size}$ vectors)
- Data transfer costs (typically minor)

Project costs at 50M, 100M, and 200M queries per month to show scalability

Calculate ROI: If latency improvement reduces churn from 8% to 6.5%, that protects ~\$900K annual revenue. What's the payback period?

Step 4: Build Weighted Decision Matrix

Define decision criteria with business-justified weights:

- Latency (40% weight): Client complaints cite speed as #1 issue
- Accuracy (35% weight): Legal search requires high precision
- Cost (15% weight): Budget flexibility but cost matters
- Implementation complexity (10% weight): Migration timeline and risk

Score each model on each criterion (1-10 scale with clear rubric)

Calculate weighted scores: $\text{model_score} = \sum(\text{criterion_score} \times \text{weight})$

Rank models by total weighted score

Step 5: Executive Recommendation and Risk Mitigation

Select top-ranked model based on decision matrix

Quantify business impact: latency improvement → user satisfaction → churn reduction → revenue protection

Document risks (e.g., 2% accuracy drop, migration complexity) and mitigation strategies

Propose monitoring plan: metrics to track (p95 latency, recall@10, error rate), alert thresholds, dashboard design

Define rollback plan: if client complaints increase $>10\%$ or recall drops $<83\%$, revert immediately

Suggest gradual rollout: 5% traffic week 1, 25% week 2, 50% week 3, 100% week 4

Compile Your Findings

Requirement 1: Benchmark Results Table

A comprehensive comparison of all four embedding models across key performance metrics. Include: Recall@5 and recall@10 accuracy scores, latency percentiles (p50, p95, p99), throughput measurements (queries per second), and MRR rankings for each model tested.

Requirement 2: Cost Model Spreadsheet

→ A detailed total cost of ownership analysis for each embedding model at production scale. Include: One-time embedding generation costs, monthly recurring compute and storage costs, projections at 50M, 100M, and 200M query volumes, and ROI calculations connecting cost to churn reduction and revenue protection.

Requirement 3: Weighted Decision Matrix

→ A structured evaluation framework reflecting LegalTech's business priorities. Include: Criteria definitions with justified weights (latency, accuracy, cost, complexity), scoring rubric for each criterion, individual model scores, weighted calculations, and final model rankings.

Requirement 4: Executive Recommendation Document

→ A business-ready recommendation for LegalTech's executive team. Include: Top model selection with supporting rationale, quantified business impact (latency improvement, churn reduction, revenue protection), identified risks with mitigation strategies, monitoring plan with alert thresholds, and phased rollout schedule.

Summary



This activity provides hands-on experience with production ML decision-making through practical application of multi-dimensional benchmarking, cost modeling, and weighted evaluation frameworks. Participants have developed competency in measuring accuracy-latency tradeoffs across embedding models, calculating total cost of ownership at enterprise scale, and translating technical metrics into executive recommendations, which are directly applicable to ML architecture decisions in production SaaS environments.

Share The Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.

Instructions for Documenting and Sharing The Project for Peer Review

1. Document the Project

→ Use word processing software to create your report.

→ Ensure all sections of the project document structure are completed.

→ Proofread and edit your report for clarity and accuracy.

2. Share the Project

→ Save your report in PDF format.

→ Upload your documents to the “Peer Review” area in the course shell.

→ Provide a brief description of your project in the submission post.

Instructions for Documenting and Sharing The Project for Peer Review

3. Peer Review

- Review the projects submitted by your peers.
- Provide constructive feedback and comments on their work.